

PHILIPS AKINBAMI

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SUMMARY

M.S. Telecommunications candidate at the University of Maryland (May 2026) with a B.S. in Electrical Engineering, focused on computer networking, protocols, and network automation. Hands-on with routing and switching, packet and traffic analysis (Wireshark, tcpdump), and automation via Ansible and NETCONF/RESTCONF, backed by strong data-analytics skills in Python, SQL, and Power BI applied to large-scale network and engineering datasets.

EDUCATION

University of Maryland

M.S. in Telecommunications

College Park, MD

May 2026

Relevant Coursework: Introduction to Digital Communications Systems, Networks and Protocols I & II, AI Applications for 6G, Cellular Communication Networks, 5G System Design and Evolution.

Howard University

B.S. in Electrical Engineering, GPA 3.70

Washington, DC

May 2024

Relevant Coursework: Wireless Communication, Communications Theory, Electromagnetics, Signals and Systems, Electronics & Solid-State Devices, Introduction to Computer Networks.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MATLAB, Pandas, NumPy, Matplotlib, Excel, Power BI, Git

Networking & Protocols: TCP/IP, IPv4/IPv6, VLANs, STP, SDN, DHCP, DNS, LLDP, SNMP, Routing & Switching

Network Automation: Ansible, NETCONF, RESTCONF, RESTful APIs

Systems & Infrastructure: Linux, Bash/Shell, VMware, Docker, Kubernetes, Terraform

Network Tools & Platforms: Wireshark, tcpdump, GNS3, Juniper Junos

Cloud & Design: AWS, AutoCAD / CAD Basics

WORK EXPERIENCE

Cummins

Columbus, IN

System Analytics and Modeling Intern

May 2023 – Aug 2023

- Created Power BI dashboards used by **20+ engineers** to monitor KPIs, improve visibility into operational performance, and support data-driven decision making.
- Processed and consolidated datasets of **~100,000+ records**, improving reporting accuracy and enabling reliable KPI tracking for engineering teams.
- Authored step-by-step documentation supporting the team's migration from MATLAB workflows to Power BI, improving data-visualization consistency.

Reliability Engineering Intern

May 2022 – Aug 2022

- Validated components using MATLAB and Excel-based reliability models across **10+ design evaluation cycles**.
- Documented failure trends and system performance issues, producing **8+ technical reports** for engineering review.
- Designed an onboarding database that streamlined training and improved workflow efficiency for a team of **15+ members**.

PROJECTS

Network Traffic Measurement & Analysis (MAWI Dataset) · Personal Project

Ongoing

- Parsed and analyzed **1M+ network packets**, identifying **6,000+ TCP retransmissions** to evaluate reliability, detect bottlenecks, and troubleshoot traffic behavior.
- Extracted protocol, retransmission, and packet-level features from captures to build structured datasets for performance analysis and debugging.
- Built Power BI dashboards to surface traffic trends, abnormal network behavior, and reliability issues.

UAV Communications · Senior Design with Raytheon Technologies

Aug 2023 – May 2024

- Improved UAV communication performance by **20%** through telemetry and ROS system optimization.
- Coordinated biweekly with **4+ Raytheon engineers** to integrate telemetry systems meeting aerospace standards.
- Validated UAV telemetry and communication subsystems across **10+ autonomous flight tests**, analyzing real-time performance, sensor accuracy, and reliability.